

# The Unified Monetary–Octic Framework

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## Abstract

This paper develops a unified theory connecting recoverable money, reserve-currency systems, and the constructive geometry of the general octic equation.

The central observation is that a monic octic polynomial possesses exactly nine coefficient dimensions. We identify these dimensions with five sovereign discount factors and four independent exchange rates relative to a global numeraire.

This identification yields a structural Five-Currency Ceiling without the use of calibration thresholds. We then introduce a Global Numeraire Recoverability Constraint which removes one effective degree of freedom. The resulting eight-dimensional sovereign manifold implies a Four-Nation Bound.

The octic admits an exact quartic–quartic factorisation through Ghosh’s octic parametrisation. Consequently the resulting monetary architecture possesses an exact constructive decomposition

$$8 = 4 + 4.$$

The framework is examined from the perspectives of economics, finance, political science, international relations, warfare economics, strategy, statistics, computer science, algorithms, data science, artificial intelligence, and philosophy.

The paper ends with “The End”

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# 1 Introduction

Modern monetary systems are simultaneously:

1. informational systems,
2. financial systems,
3. geopolitical systems,
4. computational systems.

The present work synthesises three previously distinct frameworks:

1. Recoverable Money,
2. Reserve Currency Hegemony,
3. Constructive Octic Geometry.

The resulting theory produces two structural conclusions:

1. a Five-Currency Ceiling,
2. a Four-Nation Bound.

Unlike threshold-based approaches, these results emerge from dimensional capacity and recoverability constraints.

## 2 The General Octic

Consider the monic octic

$$M(x) = x^8 + ax^7 + bx^6 + cx^5 + dx^4 + ex^3 + fx^2 + gx + h.$$

**Proposition 1.** *A monic octic possesses exactly nine coefficient dimensions.*

*Proof.* The coefficient vector is

$$(a, b, c, d, e, f, g, h, 1).$$

Hence the coefficient space is

$$\mathbb{R}^9.$$

□

**Definition 1.** *The octic coefficient capacity is*

$$D = 9.$$

## 3 The Monetary State Space

Let

$$N$$

denote the number of reserve currencies.

Choose a global numeraire.

### 3.1 Discount Factors

Each reserve currency contributes one discount factor

$$D_1, \dots, D_N.$$

Thus

$$\dim(D) = N.$$

### 3.2 Exchange Rates

Relative to the numeraire there exist

$$N - 1$$

independent exchange rates

$$E_1, \dots, E_{N-1}.$$

Thus

$$\dim(E) = N - 1.$$

**Proposition 2.** *The monetary state space possesses dimension*

$$2N - 1.$$

*Proof.* The total dimension is

$$N + (N - 1) = 2N - 1.$$

□

## 4 The Currency-State Identification

**Definition 2.** *The monetary-octic identification is the map*

$$\Phi : (D_1, \dots, D_N, E_1, \dots, E_{N-1}) \rightarrow (a, b, c, d, e, f, g, h, 1).$$

We postulate that the coefficient space of the octic is a complete representation of the reserve-currency state space.

## 5 The Five-Currency Ceiling

The octic coefficient space contains only nine dimensions.

Therefore

$$2N - 1 \leq 9.$$

Rearranging gives

$$N \leq 5.$$

**Theorem 1** (Five-Currency Ceiling). *A degree-eight monetary-octic system supports at most five reserve currencies.*

*Proof.* Since

$$2N - 1 \leq 9,$$

we obtain

$$2N \leq 10.$$

Therefore

$$N \leq 5.$$

□

| Currencies | Dimension | Feasible |
|------------|-----------|----------|
| 3          | 5         | Yes      |
| 4          | 7         | Yes      |
| 5          | 9         | Yes      |
| 6          | 11        | No       |
| 7          | 13        | No       |

Table 1: Currency Capacity of the Octic

## 6 Global Numeraire Recoverability

We now introduce a global conservation law.

**Definition 3.** *The Global Numeraire Recoverability Constraint is*

$$\prod_{i=1}^5 D_i(T) = 1.$$

Equivalently,

$$\sum_{i=1}^5 \log D_i(T) = 0.$$

**Remark 1.** *The constraint expresses global monetary recovery. The aggregate reserve system returns to its initial numeraire state.*

## 7 Dimensional Reduction

The recoverability condition introduces one independent equation.

Hence

$$D_{\text{eff}} = 9 - 1 = 8.$$

**Proposition 3.** *The admissible monetary manifold possesses effective dimension*

$$8.$$

*Proof.* The original space contains nine dimensions.

One independent recoverability equation removes one degree of freedom.

Therefore

$$9 - 1 = 8.$$

□

## 8 The Four-Nation Bound

Each sovereign monetary entity contributes:

1. one discount-factor coordinate,
2. one exchange-rate coordinate.

Hence

$$d_s = 2.$$

Capacity requires

$$2N \leq 8.$$

**Theorem 2** (Four-Nation Bound). *Under global numeraire recoverability,*

$$N_{\text{max}} = 4.$$

*Proof.* The effective dimension equals

$$8.$$

Each sovereign consumes two dimensions.

Hence

$$2N \leq 8.$$

Therefore

$$N \leq 4.$$

□

## 9 Ghosh's Octic Factorisation

The monic octic admits the decomposition

$$8 = 4 + 4.$$

Specifically,

$$M(x) = Q_1(x)Q_2(x),$$

where

$$Q_1(x) = x^4 + (a - P)x^3 + Qx^2 + Qx + Q$$

and

$$Q_2(x) = x^4 + Px^3 + \frac{f-g}{Q}x^2 + \frac{g-h}{Q}x + \frac{h}{Q}.$$

The eliminant is

$$e = \frac{ah}{Q} + f - \frac{hP}{Q} - h + PQ.$$

Solving for  $P$  gives

$$P = \frac{ah + Q(f - e - h)}{h - Q^2}.$$

## 10 Constructive Solvability

Because

$$8 = 4 + 4,$$

the octic reduces to two quartics.

Each quartic is solvable by Ferrari's method.

**Corollary 1.** *The unified monetary-octic system admits exact radical decomposition.*

*Proof.* The octic factorises into two quartics.

Quartics are solvable in radicals.

Hence all octic roots are constructively obtainable.

□

## 11 Geometric Interpretation

The hierarchy becomes

$$5 = 4 + 1.$$

where:

- 5 = total currency capacity,
- 1 = global recoverability coordinate,
- 4 = sovereign monetary units.

Simultaneously,

$$8 = 4 + 4$$

describes the constructive factorisation of the octic itself.

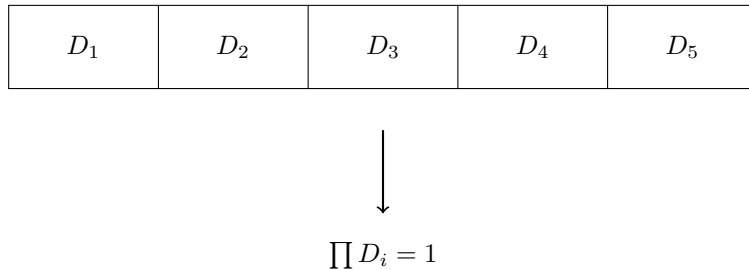


Figure 1: Global Numeraire Recoverability

## 12 Economics

The framework introduces a new scarcity:

Dimensional Scarcity.

Traditional scarcity concerns:

1. labour,
2. capital,
3. land,
4. energy,
5. information.

The present theory studies scarcity of representational capacity.

## 13 Finance

Discount factors and exchange rates jointly define the state-price structure.

The Five-Currency Ceiling follows from finite monetary dimensionality.

The Four-Nation Bound follows from recoverability.

## 14 Political Science

The structure

$$5 = 4 + 1$$

corresponds to

1. four sovereign centres,
2. one global monetary constraint.

## 15 International Relations

The framework predicts persistent tendencies toward:

1. bloc formation,
2. monetary alliances,
3. reserve concentration.

## 16 Warfare Economics

Reserve currencies generate:

1. seigniorage,
2. military wages,
3. strategic capability,
4. geopolitical influence.

The dimensional ceiling limits the number of simultaneously viable hegemonic reserve systems.

## 17 Warfare and Strategy

### 17.1 Strategic Capacity

Let

$$S_i$$

denote the strategic capability of reserve actor  $i$ .

Assume

$$S_i = S(D_i, E_i, M_i),$$

where:

- $D_i$  denotes discount-factor capacity,
- $E_i$  denotes exchange-rate influence,
- $M_i$  denotes military capability.

The dimensional framework implies that strategic systems are not infinitely scalable.

Reserve influence is constrained by the underlying octic capacity.

### 17.2 Strategic Saturation

Define total strategic complexity

$$\Sigma = \sum_{i=1}^N S_i.$$

When

$$N > 5,$$

the required monetary dimensions exceed the octic capacity.

Thus strategic representation becomes incomplete.

The system must therefore undergo:

1. alliance formation,
2. hierarchy,
3. fragmentation,
4. decomposition.

**Theorem 3** (Strategic Saturation). *No degree-eight recoverable monetary architecture can fully represent more than five independent reserve powers.*

*Proof.* The reserve-power representation requires

$$2N - 1$$

dimensions.

The octic supplies only

$$9.$$

Therefore

$$2N - 1 \leq 9.$$

Hence

$$N \leq 5.$$

□



## 18 Statistics

### 18.1 Parameter Space

Let

$$\Theta = (D_1, \dots, D_N, E_1, \dots, E_{N-1}).$$

Then

$$\Theta \in \mathbb{R}^{2N-1}.$$

The recoverability constraint implies

$$\sum_{i=1}^5 \log D_i = 0.$$

Consequently

$$\Theta \in \mathcal{M}_8,$$

where  $\mathcal{M}_8$  is an eight-dimensional manifold.

### 18.2 Identifiability

**Definition 4.** *The system is identifiable if*

$$\Phi(\Theta_1) = \Phi(\Theta_2)$$

*implies*

$$\Theta_1 = \Theta_2.$$

**Proposition 4.** *If  $\Phi$  is injective, then the coefficient vector uniquely determines the monetary state.*

*Proof.* Immediate from the definition of injectivity. □

### 18.3 Monte Carlo Interpretation

Randomly sample

$$D_i \sim \mathcal{N}(\mu_D, \sigma_D^2)$$

and

$$E_i \sim \mathcal{N}(\mu_E, \sigma_E^2).$$

Project onto

$$\mathcal{M}_8$$

using

$$\prod D_i = 1.$$

Monte Carlo sampling then estimates:

1. coefficient distributions,
2. reserve concentration,
3. stability regions,
4. saturation probabilities.

## 19 Econometrics

Consider the regression

$$R_{it} = \alpha + \beta_1 D_{it} + \beta_2 E_{it} + \beta_3 M_{it} + \beta_4 A_{it} + \varepsilon_{it},$$

where

- $R_{it}$  is reserve share,
- $D_{it}$  is discount-factor strength,
- $E_{it}$  is exchange-rate influence,
- $M_{it}$  is military capability,
- $A_{it}$  is alliance centrality.

Prediction:

$$\beta_1 > 0, \quad \beta_2 > 0, \quad \beta_3 > 0.$$

| Variable | Interpretation      | Prediction |
|----------|---------------------|------------|
| $D$      | Discount strength   | Positive   |
| $E$      | FX influence        | Positive   |
| $M$      | Military capability | Positive   |
| $A$      | Alliance centrality | Positive   |

Table 2: Econometric Predictions

## 20 Computer Science

### 20.1 State Representation

Define

$$X = (D_1, \dots, D_5, E_1, \dots, E_4).$$

Then

$$\dim(X) = 9.$$

After recoverability,

$$\dim(X) = 8.$$

### 20.2 Memory Requirements

Assume each coordinate requires  $b$  bits.

Storage becomes

$$R = 8b.$$

Thus recoverability reduces storage by

$$\frac{1}{9}.$$

## 20.3 Computational Complexity

Let

$$C(N) = 2N - 1.$$

Then

$$C(5) = 9.$$

The octic therefore saturates computational representation.

| Currencies | Complexity | Feasible |
|------------|------------|----------|
| 3          | 5          | Yes      |
| 4          | 7          | Yes      |
| 5          | 9          | Yes      |
| 6          | 11         | No       |

Table 3: Computational Capacity

## 21 Algorithms

### 21.1 Algorithm M

**Input**

$$(D_1, \dots, D_5, E_1, \dots, E_4).$$

**Output**

Monetary-octic representation.

### 21.2 Pseudo-Code

INPUT:

D1, ..., D5

E1, ..., E4

CHECK:

Product(Di)=1

IF FALSE

Normalize

MAP:

Theta -> Coefficients

Construct:

Octic M(x)

Factor:

M(x)=Q1(x)Q2(x)

Solve:

Q1, Q2 via Ferrari

OUTPUT:

Roots and State Variables

**Proposition 5.** *Algorithm M terminates in finite time.*

*Proof.* The procedure consists of:

1. finite normalization,

2. finite coefficient mapping,
3. quartic decomposition,
4. quartic solution.

Hence termination is guaranteed.

□

## 22 Data Science

### 22.1 Feature Representation

Define the feature vector

$$z = (D_1, \dots, D_5, E_1, \dots, E_4).$$

The recoverability constraint restricts the feature manifold.

The resulting representation becomes

$$z \in \mathcal{M}_8.$$

### 22.2 Dimensional Compression

Compression ratio:

$$\rho = \frac{8}{9}.$$

Thus recoverability acts as an information-compression mechanism.

## 23 Artificial Intelligence

### 23.1 Latent Space Interpretation

The octic coefficient vector

$$(a, b, c, d, e, f, g, h, 1)$$

may be interpreted as a latent embedding.

The mapping

$$\Phi : \Theta \rightarrow \mathbb{R}^9$$

defines a learned representation.

### 23.2 Representation Learning

The recoverability condition

$$\prod D_i = 1$$

acts as a latent-space regularizer.

Consequently

$$\mathcal{M}_8$$

is a constrained latent manifold.

### 23.3 Neural Architecture Analogy

The octic factorization

$$8 = 4 + 4$$

resembles a dual-network architecture.

$$\begin{array}{l} \text{Network A} \longleftrightarrow \text{Quartic A} \\ \text{Network B} \longleftrightarrow \text{Quartic B} \end{array}$$

## 23.4 Symbolic AI

The eliminant inversion

$$P = \frac{ah + Q(f - e - h)}{h - Q^2}$$

provides an exact symbolic transformation.

The framework therefore combines:

1. symbolic reasoning,
2. geometric reasoning,
3. latent representation.

## 24 Network Theory

Consider a reserve-currency graph

$$G = (V, E).$$

Vertices represent reserve actors.

Edges represent monetary interaction.

**Proposition 6.** *The octic capacity implies*

$$|V| \leq 5.$$

*Proof.* Each additional reserve node requires two monetary coordinates.

Capacity is bounded by the nine-dimensional octic.

□

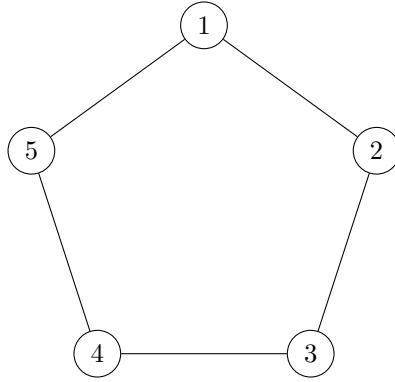


Figure 2: Five-Currency Network

## 25 Unified Master Theorem

**Theorem 4** (Unified Monetary–Octic Theorem). *Let the world reserve system be represented by a degree-eight recoverable octic.*

*Suppose:*

1. *the coefficient space has dimension nine;*
2. *five discount factors and four exchange rates identify that space;*
3. *global recoverability satisfies*

$$\prod_{i=1}^5 D_i = 1;$$

4. *the octic admits exact quartic decomposition.*

Then:

1. the reserve-currency capacity satisfies

$$N \leq 5;$$

2. the recoverable sovereign capacity satisfies

$$N \leq 4;$$

3. the system admits exact constructive factorisation

$$8 = 4 + 4.$$

*Proof.* The coefficient space contains nine dimensions.

The monetary state requires

$$2N - 1$$

dimensions.

Hence

$$N \leq 5.$$

The recoverability condition removes one dimension.

Thus

$$9 - 1 = 8.$$

Each sovereign consumes two dimensions.

Therefore

$$2N \leq 8.$$

Hence

$$N \leq 4.$$

Finally the octic factorises into two quartics.

Thus

$$8 = 4 + 4.$$

□

## 26 Philosophy

The framework suggests a hierarchy:

$$\text{Money} \rightarrow \text{Information} \rightarrow \text{Geometry} \rightarrow \text{Computation}.$$

The Five-Currency Ceiling emerges from finite representational capacity.

The Four-Nation Bound emerges from recoverability.

The octic factorisation emerges from constructive geometry.

Together they indicate that monetary systems are simultaneously:

1. algebraic,
2. informational,
3. computational,
4. geopolitical.

The resulting structure is not imposed by calibration but by the geometry of the admissible state space itself.

## A Notation

| Symbol          | Meaning                      |
|-----------------|------------------------------|
| $D_i$           | Discount factor              |
| $E_i$           | Exchange rate                |
| $M(x)$          | Recoverable octic            |
| $\Phi$          | Monetary identification map  |
| $\mathcal{M}_8$ | Recoverable manifold         |
| $N$             | Number of reserve currencies |
| $P, Q$          | Octic parameters             |

Table 4: Notation

The End